

Non-Fungible Tokens (NFTs): A Review of Pricing Determinants, Applications and Opportunities

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Abstract

This paper provides a review of the development of the non-fungible tokens (NFTs) market, with a particular focus on its pricing determinants, its current applications, and its future opportunities. We investigate the current state of the NFT markets and highlight the perception and expectations of investors towards these products. We summarize and compare the financial and econometric models that have been used in the literature for the pricing of non-fungible tokens with a special focus on their predictive performance. We design a framework that can help to understand the price formation of NFTs. We further aim to shed light on the value creating determinants of NFTs in order to better understand investors' behavior on the blockchain.

Keywords— Digital Assets, NFT, Pricing, Valuation

JEL— G12, G18, G41

1 Introduction

Non-fungible tokens (NFTs) are a novel form of unique digital asset that have emerged with the advent of blockchain technology and the rise of Web3. These digital tokens are secured by cryptography methods and recorded in smart contracts, which are tamper-proof and self-enforcing digital agreements that automatically execute when predetermined conditions are met (Cong & He, 2019). Unlike cryptocurrencies such as Bitcoin, NFTs are non-fungible and cannot be exchanged on a one-to-one basis with other tokens, as they represent a unique item (Howell et al., 2020). NFTs have the potential to revolutionize the way we think about the valuation and ownership of digital assets in the Web3 era. Recently, financial interest in NFTs has grown exponentially, capturing the attention of investors and the general public alike. As Figure 1 shows, the cumulative USD traded NFT volume skyrocketed from a modest 1.5 billion dollars in January 2019 to a staggering 180.8 billion dollars in December 2022.

[Figure 1 about here.]

This surge in popularity is attributed to their potential to disrupt traditional asset classes and reveal new investment opportunities. Financial investors' expectations are influenced by the inherent value proposition offered by NFTs, including verifiable scarcity driving demand and value, blockchain technology ensuring provenance and authenticity, clear transferable ownership rights facilitating frictionless transactions and trading, interoperability across platforms and applications, and enhanced liquidity due to the development of specialized marketplaces and exchanges (Wang et al., 2021). As a result, NFTs are poised to leave a lasting impact on the financial landscape, prompting investors to reevaluate their investment strategies and adapt to the rapidly changing digital economy.

This study examines the factors that drive the pricing of NFTs and the role of investor beliefs in shaping this rapidly emerging market. Despite the increasing and potential impact of NFTs on the financial landscape, the determinants of NFT pricing remain relatively unexplored, leaving market participants with limited guidance on how to navigate this novel asset class. In asset pricing,

researchers have long sought to identify the key drivers of asset values, including the influence of investor sentiment, macroeconomic factors, and other systematic or idiosyncratic risks (Campbell et al., 2022; Davis and Segal, 2022). As the market for NFTs continues to expand, it is essential to apply similar rigorous analyses to uncover the underlying pricing determinants and mechanisms that govern the NFT market. By doing so, we can better understand the sources of value creation and provide more informed insights to investors seeking to capitalize on the opportunities presented by NFTs.

The study of NFT pricing determinants can also help to unravel the complex interplay between investor expectations and market dynamics. In traditional asset markets, those forces play a crucial role in shaping market prices, as they are often driven by perceptions of risk, return, and the future prospects of an asset (Ozoguz, 2009; Benamar et al., 2021). In the context of NFTs, understanding how these expectations are formed and influenced by various factors can provide valuable insights into the psychology and decision-making processes of market participants. This, in turn, can help investors to distinguish between the systematic and emotional components of their investment choices, enabling them to make more informed and rational decisions when allocating resources to this emerging asset class. Our study synthesizes the existing body of research on NFT pricing determinants by drawing parallels with established asset pricing theories and models in the field. Through this comprehensive review of the existing literature, we aim to identify the gaps in our current understanding of NFT valuation and provide a solid foundation for future research in this field.

Our paper will focus on three main objectives. First, we will trace the history and motivations behind the emergence of NFTs, in order to contextualize their growth and development within the broader financial ecosystem. Second, we will examine the magnitude of NFTs' impact on markets and investors, emphasizing the extent to which this disruptive technology has captured the attention and imagination of market participants. Third, we will analyse the determinants of NFT pricing dynamics, with a specific focus on the role of cryptocurrency trends and market participants networks in shaping market outcomes. By comparing the findings from various studies in this

area, we aim to establish a clearer understanding of the factors that contribute to the so-called hype surrounding NFTs and the extent to which this sentiment influences market dynamics.

The remainder of this paper is organized as follows. Section 2 investigates the NFT ecosystem by first deepening their evolution on the blockchain and then describing the properties and characteristics of 5 different key categories related to those tokens: Gaming, Collectibles, Utility, Art, and Web3 NFTs. Section 3 discusses the current implementations of NFT pricing models, implications for market participants and outlines interesting areas for future research. Section 4 concludes and discusses opportunities within the NFT market.

2 Exploring the NFT Chronicle: Categories and Properties

The advent of blockchain technology, with Bitcoin as its pioneer, paved the way for innovative digital assets such as NFTs. Indeed, it has come a long way since 2012, when the first discussions on the possibility of reducing fungibility on one's Bitcoins were held. Specifically, in the archetypal model of Colored Coins, each Bitcoin (by nature fungible) could be distinguished from all the others by using a specific “mark” assigned through additional lines of code. Those Colored Coins could be used for specific purposes, such as creating digital assets on top of Bitcoin blockchain by using its functionalities beyond a digital currency (Crosby et al., 2016). Colored Coins allowed for further experimentation and created the prerequisites for the birth of NFTs on the Ethereum blockchain.

Launched in 2016, Ethereum enabled more advanced functionalities, especially for the possibility of implementing smart contracts. Its flexibility allows users to exploit these contracts for the creation of different assets (i.e., tokens) by the so-called Ethereum Request for Comments (ERC). These are sets of rules, or standards, which are shared and agreed by the Ethereum community and which summarize the guidelines on how a certain token must be built for it to function correctly. Then in 2017, Axiom Zen, a Canadian start-up, proposed the ERC-721 standard for writing NFTs on Ethereum. This protocol provides that the two main components of an NFT

are kept in two opposite systems: the object of the token (the image, the sound and in general all the graphic characteristics of the NFT) are stored on a private and centralized server of the deployer of the token, while the proof of ownership is stored on the blockchain. The year 2017 also saw the birth of two highly successful NFT collections: CryptoKitties, by Axiom Zen itself, and CryptoPunks, by Larva Labs.

Although Ethereum is very popular and widely used, some platforms have decided to create their own blockchain to offer users and investors a different alternative to Ethereum and, above all, to take advantage of a non-congested ledger. At the end of 2017, the lack of scalability of Ethereum was evident at the time of the launch of the Generation 0 (Gen 0) of CryptoKitties when, on the blockchain, there was an increase in pending transactions equal to 6 times. Investors, well-aware of the scarcity and sensing the possibility of very profitable future transactions, had landed in large numbers on Ethereum, causing a general slowdown of the system. In the context of asset pricing, the scarcity factor has been shown to play a critical role in the valuation of non-fungible objects, such as art or collectibles (Burton and Jacobsen, 1999; Mandel, 2009): the fewer specimen exist in circulation, the more that product is perceived as exclusive, and therefore, of value.

Figure 2 shows the number of sales that took place between January 2018 and December 2022 for the Gen 0 of CryptoKitties vs. their average price in USD. We observe that the dynamics of the number of sales and average value in USD of each Gen. 0 CryptoKitties are inverse. More specifically, at the beginning of the observation period, sales reached 4,500, which was an astonishing number for a single NFT collection at that time, but the average price for each of them was very low, barely touching 200 USD per token. We also note a substantial decline in sales after October 2018. At the end of 2018, the algorithm produced about 35,000 Gen 0 CryptoKitties before being stopped once again by the rule established by Axiom. For this reason, transactions involving this generation slowed down as users engaged in pairing their CryptoKitties to create successive generation tokens, which the market then focused on.

[Figure 2 about here.]

However, scarcity is not sufficient to preserve the value of these tokens in the long run. The lack of a fundamental value anchor for NFTs makes it difficult for investors to determine their true worth. This is because NFTs do not have an intrinsic value or cash flow that can be used to estimate their fair value, as is the case with traditional financial assets such as stocks and bonds. Instead, the value of an NFT is derived from its perceived uniqueness, scarcity, and demand among buyers. The subjective nature of these factors means that the value of NFTs can be highly volatile and subject to sudden shifts in market sentiment. This is particularly true in the absence of standardization and regulation in the NFT market, which can exacerbate price volatility and liquidity risks for investors. In the asset pricing literature, the lack of a fundamental value anchor for an asset is known as a speculative bubble. When investors base their investment decisions solely on the expectation that the price of an asset will continue to rise in the future, without regard for its underlying value, it can create an unsustainable bubble that is vulnerable to a sudden collapse. This phenomenon is often associated with irrational exuberance and the greater fool approach (Harrison and Kreps, 1978; Baker et al., 2012), which refers to the belief that one can profit from an overpriced asset by selling it to a "greater fool" in the future.

In the NFT market, it is possible to trace a potential dynamic of this greater fool type: After a period of strong enthusiasm due to the introduction of this new digital asset, sales of NFTs dropped dramatically towards the end of 2021. For instance, Jack Dorsey's (CEO of Twitter) first tweet NFT was auctioned at USD 2.9 million USD in March 2021, went up for resale at auction in April 2022 and the highest bid was just under 10,000 USD, a dramatic loss in its value of over 99 percent in just one year. Therefore, investors in the NFT market need to exercise caution and carefully evaluate the risks and uncertainties associated with investing in this new asset class.

It is important to adopt a prudent investment approach that is based on sound asset pricing principles, such as diversification, risk management, and a long-term investment perspective. Additionally, the development of standardization and regulation in the NFT market may help to provide a more stable and transparent framework for NFT valuation and trading, thereby reducing the risks and uncertainties for investors. From a capital markets perspective, NFTs represent a new

form of digital assets that can be leveraged for various purposes, including digital art, virtual real estate, and intellectual property. As the market matures, factors such as utility, rarity, and network effects may influence NFT valuations, alongside more traditional asset pricing factors like cash flows and risk premiums. Moreover, the emergence of alternative blockchains and advancements in blockchain technology may affect the NFT market dynamics, reducing congestion issues and transaction costs. This, in turn, could influence investor behavior and market efficiency, as well as the adoption of NFTs in traditional financial markets.

With 2020 and the outbreak of the Covid-19 pandemic, NFTs began to take hold among the general population, attracting investors and stimulating artists to create their own personal series. Numerous marketplaces emerged to cater to the diverse needs of NFT enthusiasts. Kireyev and Evans (2021) provide a classification of these marketplaces into two distinct categories, based on the scope of services offered and the types of tokens dealt with. The first category, referred to as streamlined marketplaces, are platforms that primarily facilitate the buying and selling of a wide variety of NFTs through auctions or direct sales, with minimal additional services. These marketplaces prioritize efficiency and speed in transactions and cater to a broad range of NFT types. Notable examples of streamlined marketplaces include OpenSea and Nifty Gateway, both of which are well-established platforms for trading digital art and collectibles. Conversely, the second category, known as augmented marketplaces, are specialized platforms that focus on specific segments of the NFT market, such as gaming, sports, or fine arts. These marketplaces offer comprehensive services that encompass the entire life cycle of an NFT, from creation to sales and marketing, including pricing recommendations and portfolio tracking. NBA's Top Shot and CryptoKitties serve as prime examples of augmented marketplaces, characterized by unique identifiers (256-bit numbers) representing the DNA and attributes of the tokens. These augmented marketplaces enable users to purchase tokens and use them within the platform to progress in the game. This creates an incentive for users to acquire increasingly rare and powerful items to outperform other players, ultimately driving demand and value for these NFTs in the marketplace.

At the beginning of 2021, NFTs gained further popularity among the general public, with a substantial value increase. In March 2021, Mike Winkelmann's (aka Beeple) NFT "Everydays: The First 5000 Days" was sold by Christie's for USD 69.3 million. This event made investors understand the profit potential of this new digital asset, thus, allowing the NFT market to grow further. Figure 2 shows a substantial increase in the average value for Gen. 0 CryptoKitties from March 2021 onwards, up to the peak of September when these tokens reached an average of USD 6,000 per NFT. However, Figure 2 also indicates the great volatility that characterizes not only the Gen. 0 of CryptoKitties, but in general the whole NFT market: From September 2021, we observe a strong decrease of about 92% of the average value per token. This was mainly caused by the dulling of the excitement and frenzy of these new digital assets. At that time, given that most of the NFTs in circulation were image collections such as CryptoKitties, investors quickly realized that they could not get any use from their possession and rushed to liquidate their positions. Finally, the news of the Russian invasion of Ukraine and the crash in the cryptocurrency markets triggered a further market contraction in 2022.

Potential future investors seeking opportunities in the NFT market should focus more on the unique aspects of this asset class. One intriguing aspect is the emotional satisfaction and utility that investors derive from their NFT holdings. Kong and Lin (2021) demonstrate that, despite the highly speculative nature of NFTs, investors are willing to tolerate significant risks to capitalize on the emotional dividends that come with owning these digital collectibles, much like their physical counterparts. In light of the recent crypto crisis, potential investors may find solace in the non-financial value that NFTs provide, adding a layer of resilience to this asset class. As NFTs represent a blend of art, culture, and technology, their appeal extends beyond conventional financial metrics, drawing a diverse range of collectors and enthusiasts who appreciate their intrinsic worth. This facet of the NFT market, combined with the increasing popularity and adoption of digital collectibles, could signal a robust and potentially profitable investment opportunity even during turbulent times in the broader cryptocurrency space. An example of this property is shown by

Figure 3, which compares the normalized values of volume and active wallets across NFT markets between January 2021 and January 2023.

[Figure 3 about here.]

The graph reveals a pronounced surge in the volume of NFTs traded towards the end of 2021, culminating in a peak at the end of January 2022, before the market experienced a sharp correction, leading to a significant decline in trading volume. Nevertheless, active wallets exhibited a more resilient trajectory, registering modest growth even during the most challenging periods for the NFT market. This divergence between trends can be attributed to several factors that go beyond mere utility. First, the persistent growth in active wallets may indicate that investors remain engaged in the NFT ecosystem, despite the fluctuations in trading volume. This enduring interest could stem from the unique attributes of NFTs, such as their ability to confer digital ownership, establish provenance, and facilitate novel forms of creative expression. Second, NFTs often possess sentimental value, which can prompt collectors to hold onto their tokens even during market downturns. This emotional attachment may further explain the sustained growth in active wallets, as it suggests that investors derive satisfaction from their NFTs beyond financial gains. Lastly, the resiliency of active wallets may be indicative of a maturing market, where participants increasingly recognize the long-term potential of NFTs and their underlying technologies. As the market stabilizes, investors may be more inclined to hold their NFTs, which can contribute to the sustained growth of active wallets.

As information is shared among investors in communities formed around NFT projects, it may contribute to the stability of active wallet trends. As shown by Han and Yang (2013), when information is exogenous, social communication can improve market efficiency in the NFT market, leading to better price discovery and potentially fostering greater confidence among investors. This, in turn, can result in a more stable and active wallet trend, as investors are more likely to engage with NFT platforms. However, when information is endogenous, social communication may cause investors to free-ride on informed peers, potentially reducing the incentive to gather new

information. In this scenario, the stability of active wallets could be influenced by the reliability of the information shared within the network, as well as the trustworthiness of informed participants.

Therefore, the role of social networks in the NFT market is twofold. On one hand, they can contribute to market efficiency and active wallet stability by facilitating information exchange and fostering investor engagement. On the other hand, they can also impact the incentives for information acquisition, which may influence the quality of information shared within the network. In summary, the role of investor networks in the NFT market is pivotal in shaping investor behavior and market dynamics. By fostering communities around NFT projects, these networks can enhance market liquidity, stimulate trading volume, and positively impact net earnings (Halim et al., 2019), as investors are more likely to stay engaged with the NFT market and continue participating in it.

In order to support investors in efficiently allocating their investment and help them to develop a comprehensive understanding of the diverse array of digital assets available in the market, it is essential to categorize NFTs. Such categorization provides investors with a clear and detailed understanding of the distinct types of NFTs and their specific characteristics, thus enabling them to make informed investment decisions. This categorization also enables investors to evaluate the potential returns and risks associated with each category, facilitating the development of standardized valuation approaches, and fostering regulatory compliance.

There is no standardized taxonomy for NFTs that has been universally adopted by the industry. Various marketplaces, industry experts, and researchers have proposed different taxonomies to classify NFTs based on their unique attributes and characteristics. The taxonomy proposed by *NonFungible.com* is one of the most widely recognized and used by researchers and practitioners alike (Nadini et al., 2021; Kong and Lin, 2021; Dowling, 2022; Boido and Aliano, 2023). One of the advantages of using *NonFungible*'s taxonomy is that it provides a clear and concise way of categorizing NFTs into five primary categories within the digital asset ecosystem. These categories include *Art*, *Collectibles*, *Gaming*, *Metaverse*, and *Utility*. By grouping NFTs into these distinct categories, it becomes easier to compare and analyse their unique characteristics, such as their

underlying value proposition, economic models, and target markets. Table 1 summarizes the five NFT categories, their key properties, and their use case project examples.

[Table 1 about here.]

In the realm of gaming, NFTs represent assets that can be utilized within video games, with their elements stored on the blockchain. This offers a significant departure from traditional video games, as players gain real ownership of in-game assets through the purchase and sale of NFTs. Gaming NFTs have demonstrated a remarkable ability to engage active users, resulting in the highest participation rates compared to other categories. This high level of user involvement translates to continuous exchanges between players, making the gaming sector highly liquid. This liquidity is further evidenced by the gaming sector attracting a substantial portion of the total USD traded in the NFT market. Examples of popular gaming NFTs include Axie Infinity, NBA Top Shot, and CryptoKitties.

Not much unlike physical collectibles, NFT collectibles are released in collections, or series, which represent variations of the same image, video or other media. The characters in the Cryptopunks project, for instance, differ from each other in certain attributes that also make the price vary: man/woman, human/alien/monkey, and presence or absence of accessories. NFT collectibles record the highest level of transactions though the number of active wallets is much lower than that of gaming NFTs. This is mainly due to two reasons: First, there are far fewer collectible marketplaces than gaming NFTs, so investors will find themselves on a single platform (e.g., OpenSea) than players who will have to migrate to different platforms for each video game in which they want to trade their assets. The second reason for this concentration of the market is due to a few large-value transactions. Nadini et al. (2021) show that the top 10% of buyer–seller pairs contribute 90 percent to the total number of NFT transactions. Examples of NFT collectibles are CryptoPunks, the Bored Ape Yacht Club (BAYC), and Azuki.

NFT utilities, the third main group, are assets that provide utility in the real or digital world through the blockchain. In other words, utility tokens give its holder consumptive rights to access a product or service (Howell et al., 2020) so that their use is not directly related to the need to collect

or play with the token of interest. In particular, because these tokens serve as the means of payment on a platform or offer access to the firm's services, they possess utility features (Gryglewicz et al., 2021). Utility NFTs comprise different categories: finance, health, supply chain, or digital ID. The most popular NFT utility projects are VeeFriends (which grant access to the VeeCon, a multi-day event exclusively for VeeFriends NFT holders), Ethereum Name Service (ENS, where users can purchase and manage domain names for their digital assets), and Nouns.

Art NFTs can be defined by exclusion from the previous sectors. Art NFTs are assets with an artistic function that have not been released in series (as could happen for collectibles) and that cannot be used within any type of video game hosted on the blockchain. This type of tokens has brought many innovations to the art market, especially due to the easing of barriers to entry this opaque market. Everyone can create and sell their works on different platforms in much shorter time than on the traditional art market, with an average time between purchase and resale in art NFTs of just 33 days versus the average resale period on the traditional art market of 25 to 30 years (McAndrew, 2023). Furthermore, art NFTs have addressed issues that have affected the traditional art market for decades, such as provenance, title, authenticity and a fairer distribution of income. The creation of communities by the artists themselves via social networks, such as Twitter gravitating around their NFTs collections, have allowed for a much deeper involvement of buyers. It is the emotional and social values combined with the economic one that make NFTs a different and interesting crypto asset, which reflects pretty much the same motivations of collectors in the traditional art market. Main examples of art NFTs are ArtBlocks, i.e., tokens representing generative art through an algorithm, SuperRare, and The Currency by artist Damien Hirst which are 10,000 NFTs corresponding to 10,000 physical artworks stored in a physical vault.

The fifth main group, Web3 or Metaverse, can be defined as an extension and grouping of the previous ones. The Metaverse is a virtual universe accessible through a computer screen, laptop, virtual reality (VR), or any other digital system. Users who access this world can create their virtual avatar and interact with the surrounding reality, including other users. They can purchase virtual plots of land within the Metaverse to create their own organizations and host events. In

many cases, firms have established virtual businesses and created a space where they can offer goods and services, promote their products and organizations, and hold virtual events (Goldberg et al., [2021](#)). Some examples are the game developer company Atari in Decentraland, Adidas in The Sandbox, and Cryptovoxels.

3 Drivers of NFTs Pricing

The categorization of the entire NFT market into these five specific segments (Art, Collectibles, Games, Utility, and Web3) is essential for the purpose of understanding the pricing determinants of NFT collections. More specifically, we can systematically analyse the distinctive characteristics and underlying value drivers within each category, thus enabling a more nuanced and accurate assessment of the market dynamics. This segmentation allows for a comprehensive understanding of the varying factors that influence the valuation of NFT collections across different use cases, thereby providing valuable insights into investor behavior and decision-making processes within the rapidly evolving NFT landscape.

We distinguish three main streams of research, each addressing different aspects of NFT valuation. The first group emphasizes the analysis of within-collection characteristics, striving to identify the determinant factors that impact NFT valuations while assessing market efficiency. Researchers within this domain scrutinize the attributes of individual NFTs, such as rarity, uniqueness, and provenance, along with factors like the creator's reputation and the level of community engagement. They seek to determine how these factors contribute to the perceived value and pricing of digital assets within a specific collection. This area of research delves into the efficiency of the NFT market, exploring aspects like price discovery, the existence of arbitrage opportunities, and the influence of information asymmetry on market dynamics, thus providing valuable insights into the functioning of NFT markets.

The second group, the between-collections stream, explores the correlations among various NFT collections and the impact of investor network effects on NFT pricing. This research direction

aims to understand the interactions between different collections, identifying potential spillover effects and cross-collection influences on asset pricing. This body of literature examines the role of network effects in driving demand and the impact of investor behaviors, preferences, and market sentiment on the valuation of digital assets across multiple collections. By investigating these relationships, researchers can gain a deeper understanding of the interconnectedness of the NFT market and the underlying forces that drive its growth and development.

Lastly, the outside-collections stream concentrates on the co-movements between the NFT market and other markets, encompassing cryptocurrencies, traditional assets, and alternative investments. This line of research seeks to comprehend the extent to which NFT valuations are influenced by broader market forces and assess the potential of NFTs as a hedge, diversifier, or risk factor within an investor's portfolio. Scholars in this area study the correlations between NFT prices and various asset classes, such as cryptocurrencies like Bitcoin and Ethereum, stocks, bonds, commodities, and other alternative investments like real estate and collectibles. This approach enables researchers to uncover insights into the macroeconomic and systemic factors that affect NFT pricing and evaluate the potential role of NFTs within the wider financial ecosystem.

3.1 Within-Collections Drivers

Kong and Lin (2021) examine the CryptoPunks collection and find that the rarity of a token and the presence of certain accessories (attributes) significantly affected its price, implying that investors are willing to pay a premium for unique or scarce NFTs. Similarly, Schaar and Kampakis (2022) study the CryptoPunks collection and conclude that rarity and the number of attributes play a crucial role in determining the token's price, reinforcing the notion that perceived scarcity and uniqueness drive value. For example, tokens with the "alien" type fetched three times the average selling price, and tokens with zero attributes were priced considerably higher than those with attributes. Goldberg et al. (2021) apply the hedonic regression model to the Decentraland project, a virtual world built on the Ethereum blockchain, and discover that the value of the digital real estate (LAND) was influenced by their location and memorability, highlighting the importance of these

factors for investors seeking to capitalize on virtual real estate opportunities. Nadini et al. (2021) conduct a comprehensive analysis of the top collections in various NFT categories and find that the price of secondary sales was strongly correlated with the price of primary sales, particularly when short estimation periods were considered, suggesting that investors could potentially benefit from understanding and predicting primary sale prices to assess secondary market opportunities. They also discover that centrality in combination with visual features could outperform other regressors, especially in metaverse NFTs, indicating that investors may need to consider both social and visual aspects when evaluating NFTs in virtual worlds.

Kireyev and Lin (2021) perform a very different study from those discussed so far as their goal is to demonstrate that traditional pricing methods can cause inaccurate valuations of NFTs. They focus on the CryptoKitties collection during 2019 and compare the R2 of a hedonic linear regression model and a gradient boosting machine (GBM), a supervised machine learning model based on ensembles of trees. They find that the GBM algorithm significantly outperforms the hedonic regression model (51% vs. 41%). They argue that the hedonic regression approach is not a suitable model for the NFT valuation space due to (i) mispricing bias: The observed sales prices reflect the sub-optimal decisions of the seller in setting the price of the NFT. In fact, the high range of prices that can be found on the market pushes users to wait a long time before completing the purchase. The result is a set of auctions that are successful only when the ending price is substantially low. Given that the hedonic regression is estimated starting from the observed sale prices, the coefficients very likely incorporate this bias; and due to (ii) selection bias: In the case of auction sales, the hedonic regression model focuses only on successful vs. unsuccessful ones, resulting in higher valuations for the NFTs sold. In other words, items with a lower selling rate tend to have higher prices, so when they do sell, they sell for a higher price. The hedonic model considers only successful sale prices and gives a higher valuation to these assets (Korteweg et al., 2016). Specifically, it yields higher valuations for items from unsuccessful auctions compared to items from successful auctions. The authors conclude that, as the ID and generation of each CryptoKitties cat increase, the price of the token itself decreases.

We argue that these value determinants are peculiar of the collectibles and gaming NFT sectors. Indeed, rarity, specific attributes, and social factors, are particularly relevant because they satisfy the utility of collecting for investors, which is driven by the desire to possess unique, scarce, and visually appealing items that hold personal or cultural significance. Collectibles, by their nature, derive much of their value from their rarity and exclusivity, as collectors often compete to acquire limited-edition or one-of-a-kind items that distinguish their collection from others. This exclusivity and rarity appeal to collectors' intrinsic motivations, such as the thrill of the hunt, the satisfaction of completing a set, or the desire to showcase their tastes and interests. Furthermore, the social aspects of collecting, including the sense of belonging to a community and the potential for recognition and status, play a crucial role in driving the value of NFTs in these sectors.

In the gaming sector, the utility of collecting is often intertwined with the functional utility of the NFTs themselves, as in-game assets like characters, weapons, or virtual real estate can provide both entertainment and a sense of accomplishment to players. Rarity and attributes in gaming NFTs are particularly important, as they can affect gameplay, enable the completion of specific objectives, or enhance the player's in-game experience. For example, a rare character with powerful attributes may give the player a competitive advantage, while a unique in-game item might offer exclusive customization options or unlock new content. The combination of functional and collectible value in gaming NFTs creates a powerful incentive for players to invest time and resources into acquiring these assets, which in turn can drive up their market value.

The rise of metaverse platforms, where users can socialize, trade, and showcase their digital possessions in virtual worlds, further highlights the importance of the determinants of value in the collectibles and gaming sectors. In these virtual environments, NFTs serve not only as digital collectibles but also as symbols of status, identity, and cultural capital. Owning a scarce or unique NFT can translate into social recognition, influence, or even monetary rewards within the metaverse, as other users may be willing to pay a premium to acquire or interact with such assets. Consequently, understanding and targeting the determinants of value in collectibles and gaming

NFTs become essential for investors who seek to capitalize on the market opportunities arising from the growing intersection of collecting, gaming, and virtual social experiences.

3.2 Between-Collections Drivers

The NFT market has been the subject of extensive research, with numerous studies examining the between-collections pricing determinants, such as the correlations among various NFT collections and the impact of investor network effects on NFT pricing. In an attempt to understand the intricate dynamics of the market, Ante (2022b) analyses 14 prominent NFT projects across gaming, collectibles, art, and metaverse categories. The authors identify significant short-run relationships between different projects, for instance, the influence of CryptoPunks on CryptoKitties and The Sandbox, among others. For instance, they find that an increase in the number of sales on younger, less popular projects resulted in a decrease in sales on larger, more popular projects. They argue that the NFT market is characterized by a high degree of novelty and innovation, with investors continually searching for new and potentially lucrative investment opportunities. As a result, when a new, less-established project starts to gain traction, it may attract attention and resources away from the more established projects, thereby reducing their sales. This shift in focus can be attributed to the inherent risk-seeking behavior of investors in the NFT space, who may be willing to take on more risk in pursuit of potentially higher returns. Another explanation is that since the NFT market is highly interconnected, investors are often participating in multiple projects and collections simultaneously. Consequently, when an investor decides to allocate more resources to a newer project, they may be forced to divest from other projects to free up capital, thereby causing a decrease in sales for the larger, more popular projects. This reallocation of resources can lead to a ripple effect, where the success of one project can have a direct impact on the performance of others.

Ante (2022b) also shows that the number of CryptoPunks sales had a positive impact on the number of tokens traded for other projects. This finding can be attributed to the influence of network effects and investor sentiment in the NFT market. As a large, popular project like

CryptoPunks experiences an increase in sales, it generates positive market sentiment and increased visibility, which can spill over to other projects. This can create a virtuous cycle, wherein the success of one project drives interest and investment in others, leading to a positive feedback loop that can benefit multiple projects simultaneously.

The research conducted by Vasan et al. (2022) on the Foundation platform provides valuable insights into the dynamics of the NFT market, highlighting the crucial role of network effects in driving the value of NFTs. Their study demonstrates that the interactions between artists and collectors are essential in shaping the market, as collectors tend to develop unique tastes in digital art, leading them to make repeated purchases from a select group of artists. As a result, high-reputation artists can secure a stable income from multiple, disproportionate investments made by a small group of collectors, while low-reputation artists face difficulties in establishing a consistent collector base. This phenomenon underscores the importance of network effects in the NFT market, as the relationships between artists and collectors significantly impact the valuation and demand for digital art.

Interestingly, this observation is consistent with research conducted on the traditional art market, where network effects also play a pivotal role (Yogev and Grund, 2012; Ertug et al., 2016; Braden and Teekens, 2019). In the conventional art market, artists' reputations, gallery representations, and relationships with collectors and art dealers can heavily influence their success and the value of their work. The connections and networks formed within the art world can lead to increased visibility, recognition, and, ultimately, higher demand for an artist's creations. Additionally, collectors with strong ties to a specific artist may contribute to the artist's success by promoting their work, driving up demand and prices in the process.

The similarities between the NFT market and the traditional art market in terms of network effects can be attributed to the human element inherent in both markets. Despite the digital nature of NFTs, the underlying motivations and behaviors of artists, collectors, and investors are still deeply rooted in the social and psychological factors that drive human interactions. People naturally form connections and exhibit loyalty towards entities they appreciate, leading to the

development of unique tastes and preferences that drive their collecting behavior (Kumar and Reinartz, 2016; Siebert et al., 2020). Consequently, the network effects observed in the NFT market are a reflection of the broader human dynamics at play in the realm of art and culture.

Oh et al. (2022) examine the performance of experienced and inexperienced investors in the NFT market, shedding light on the various factors that contribute to their success or failure in trading digital assets. The authors find that experienced investors outperformed inexperienced investors by approximately 8.6 percentage points per trade, primarily due to their propensity to participate in primary markets during the mint stage, where expected returns are systematically higher. The study shows that the outperformance of experienced investors in secondary markets was largely driven by superior trade execution rather than collection-picking or market-timing abilities. These findings have implications for understanding the frictions and barriers present in the NFT market, as well as offering insights into designing effective NFT investment strategies.

The findings by Oh et al. (2022) further highlight the differences in average returns between primary and secondary market NFT trades, suggesting that price-setting mechanisms in the NFT primary markets may be inefficient, or that investors require a premium for investing in risky NFT collections. This observation aligns with the findings of the physical art market (Goetzmann et al., 2011; Lovo and Spaenjers, 2018; Whitaker and Kräussl, 2020; Kleinhans and LaTour, 2022; Jensen, 2022): Primary market sales often represent opportunities for collectors and investors to acquire artwork at lower prices before they appreciate in value, and where the risk associated with investing in new, unproven artists can contribute to a pricing premium.

3.3 Outside-Collections Drivers

In their comprehensive study on NFT pricing dynamics, Borri et al. (2022) construct an extensive data set encompassing diverse NFT categories and collections spanning from the period of early 2018 to late 2021. Their findings that the NFT market index accounts for a significant portion of NFT return variations, and that collection-specific indices exhibit even higher explanatory power, suggest that there are common factors driving NFT returns, as well as collection-specific

factors that impact returns within a particular collection. The use of collection-specific indices could aid in the development of investment strategies that target specific segments of the NFT market, potentially allowing for better risk management and diversification opportunities. The authors further examine the interplay between the NFT market and factors derived from other asset markets, including cryptocurrencies, equities, commodities, and currencies. Their findings reveal a significant exposure of the NFT market excess return to the cryptocurrency market return and a predictive capacity for future cumulative excess returns of NFT-linked cryptocurrencies.

Borri et al. (2022) also investigate the cross-sectional predictability of individual NFT returns. They identify a size effect in the NFT market reminiscent of the well-established effect in the equity market, as documented by Banz (1981). The authors also uncover that NFTs with elevated average weekly returns in prior repeat sales typically underperformed their corresponding NFT market index. This reversal effect aligns with the long-term reversal effect observed in the equity market by De Bondt and Thaler (1985) and, as suggested by Fama and French (1993) and Asness et al. (2013), is connected to the value effect: Past strong performance of an NFT does not guarantee continued success in the future and might even indicate a higher likelihood of underperformance. This finding suggests that investors should look beyond past performance when assessing the potential value of an NFT investment. Instead, they should consider a broader range of factors, such as the underlying fundamentals, market trends, and the overall quality and uniqueness of the NFT, to make more informed decisions. By doing so, investors can better manage risk and avoid basing their investment strategies on historical returns alone, which may not be a reliable indicator of future performance in the NFT market.

Ante (2022a) explores the relationship between the sales of non-fungible tokens (NFTs) and the prices of two major cryptocurrencies, Bitcoin (BTC) and Ether (ETH) by analysing the NFT transactions that took place on the Ethereum blockchain between January 2018 and May 2021. The author finds significant cointegration between four variables: NFT sales, NFT wallets (i.e., the number of unique digital wallets holding NFTs), and the prices of BTC and ETH, respectively. He

concludes that these variables share a long-run relationship, suggesting that changes in the prices of BTC and ETH have an influence on NFT sales and wallets.

Building on this research, Dowling (2022) aims to understand the correlation between the NFT markets and ETH more specifically. The author employs wavelet analysis, which is a powerful mathematical tool used to study the correlations and interactions between time series data at different frequencies, and focuses on three prominent NFT markets: CryptoPunks, Axie Infinity, and Decentraland, using these as a proxy for the broader NFT market. He finds positive short-term correlations between ETH and all three NFT markets over the period from March 2019 to March 2021. During the short-term, NFT prices tended to move in the same direction as the price of ETH, revealing a connection between the NFT market and the cryptocurrency market.

The observed correlation between NFTs and leading cryptocurrencies such as BTC and ETH can be ascribed to multiple underlying factors. Primarily, NFTs are predominantly built on the Ethereum blockchain, making their value and utility closely tied to the performance and adoption of the Ethereum network. Furthermore, the acquisition, disposal, and exchange of NFTs frequently necessitate the utilization of cryptocurrencies as the principal transaction medium, thereby establishing an immediate connection between NFT market operations and the demand for cryptocurrencies. Moreover, both NFTs and cryptocurrencies belong to the overarching digital asset ecosystem, characterized by similar investor profiles, technological frameworks, and market sentiment. Further research should investigate how fluctuations in the cryptocurrency domain can influence the perception of NFTs as a feasible investment alternative and vice versa, culminating in the detected correlation between their respective valuations.

Prior to the COVID-19 pandemic, NFTs displayed a relatively independent behavior from other asset classes, with their prices predominantly influenced by BTC and stocks, while outpacing gold. The underlying reasons for this can be attributed to the unique characteristics of NFTs, such as their indivisibility and digital scarcity, which set them apart from traditional financial assets. However, during the pandemic, NFTs experienced a significant shift in their relationship with alternative asset classes, as reported by Umar et al. (2022). They observe stronger co-movements

between NFTs and other assets such as bonds and crude oil, indicating a change in the NFT market dynamics.

Concurrently, Aharon and Demir (2022) find that during the pandemic, NFTs became recipients of risk, exhibiting behavior similar to safe-haven assets like gold and the U.S. dollar. This shift in correlation between NFTs and other assets can be ascribed to investors seeking alternative investment opportunities that offer potentially higher returns and lower correlations to traditional financial assets during periods of market uncertainty and turmoil. The pandemic-induced economic downturn pushed investors to diversify their portfolios, resulting in a surge of interest in NFTs and other unconventional investment vehicles.

The critical role played by the pandemic in driving the NFT market growth is reminiscent of the boom other asset classes have experienced during previous catastrophic events in history, for instance, the growth seen in real estate prices after the Great Depression. These events often lead to shifts in investor behavior and the search for new investment opportunities. Researchers may explore in future studies if the speed at which asset prices move following catastrophic events today differs from the past, taking into consideration factors such as technological advancements, the rapid dissemination of information, and increased inter-connectedness between investors. By examining these aspects, researchers could gain valuable insights into how the NFT market and other asset classes might evolve and respond to future crises, ultimately informing investment strategies and risk management practices in an ever-changing financial landscape.

3.4 Pitfalls in the NFT Pricing Literature

The majority of the studies discussed above focus on individual collections (Kireyev and Lin, 2021; Goldberg et al., 2021; Schaar and Kampakis, 2022), while only a handful explore broader categories (Vasan et al., 2022) or the entire market (Nadini et al., 2021; Borri et al., 2022). This narrow approach presents several drawbacks from an asset pricing perspective, particularly when compared to the research conducted in the physical art market, where market indices are commonly used to gauge performance and trends (Mei and Moses, 2002; Renneboog and Spaenjers, 2013;

Korteweg et al., 2016). Drawing parallels between these two contexts, it becomes evident that the development of more comprehensive NFT indices is crucial for fostering a more robust understanding of the NFT market and its investment potential. Moreover, it would promote greater consistency and comparability across studies, ultimately contributing to a deeper collective understanding of the NFT market and its investment implications.

Asset pricing theory underscores the importance of systematic risk factors and diversification when evaluating investment opportunities (De Santis and Gerard, 1997; Das and Uppal, 2004; Acharya et al., 2012; Cong et al., 2021). By concentrating on specific collections, current NFT research may inadvertently overlook the broader dynamics at play, including potential correlations, co-movements, and shared risk factors across various NFT categories or the entire market. Such a limited perspective could lead to an incomplete or even distorted understanding of the true investment opportunities and risk profiles associated with NFTs, thereby impeding the development of comprehensive asset pricing models tailored to this emerging asset class. Future research should endeavor to adopt a more holistic approach by examining entire NFT categories or the market as a whole. Such an approach would not only provide a more comprehensive view of the NFT investment landscape but also facilitate the identification of shared risk factors and return drivers, which are integral components of robust asset pricing models.

Lastly, we encourage researchers to explore the application of advanced statistical and econometric techniques in their investigations, such as multivariate analysis, machine learning, and network-based approaches. Already studied in the empirical asset pricing domain (Gu et al., 2020), these methodologies have the potential to uncover complex relationships, patterns, and structures within the NFT market that may not be readily discernible through traditional univariate or descriptive analyses. By leveraging these sophisticated tools, future research can contribute to the development of more robust and reliable asset pricing models for NFTs, ultimately enhancing the investment decision-making process within this rapidly evolving domain.

4 Conclusions

We have delved into the world of Non-Fungible Tokens (NFTs) to understand the past history of the NFT market and its future growth dynamics. Our investigation encompasses a thorough examination of various NFT pricing and valuation methodologies, shedding light on their respective strengths and weaknesses. We have identified a diverse array of valuation approaches for NFTs, ranging from traditional models, such as the hedonic approach, to more innovative techniques that incorporate elements like rarity, visual features, and social sentiment. We found that NFTs possess an investment profile similar to that of classic collectibles or alternative investments, which are characterized by high yields coupled with high-risk profiles. While this might deter some investors, others may perceive this as an opportunity to diversify their portfolios and potentially achieve substantial returns.

So what is it that makes NFTs so popular with investors? We argue that there are on and off blockchain reasons. The "on blockchain" reasons concern the existence of the blockchain itself: the novelty and potential introduced by this virtual world have brought many people closer to the NFT world. Numerous research papers have shown the existence of co-movements between BTC, ETH and NFTs. It is likely that many users have decided to transform their wealth from cryptocurrencies to an investment in NFTs, some for mere speculative reasons, and some for passion towards a particular creator. Instead, the "off blockchain" reasons contributing to the popularity of NFTs encompass the extensive media coverage given to this new digital world, particularly during the year 2021. The emergence of virtual communities on social networks like Twitter and Discord, centered around specific NFT collections and their creators, has fostered a sense of belonging among users. Many are motivated to acquire a piece from a particular collection to become part of the group and feel closer to the creators. This sense of community, coupled with high-profile media coverage of NFT sales has amplified the attention and interest in this new form of investment. As a result, the NFT market has experienced rapid growth in recent years, with more people recognizing its potential as an alternative asset class.

It is important to recognize that we are still in the early stages of understanding the full potential and impact of NFTs as an investment class. Considering that blockchain technology only gained popularity in 2008 with the publication of Satoshi Nakamoto's paper, and NFTs started gaining mainstream traction with CryptoPunks in 2017, it becomes evident that this market is still in its nascent phase and has yet to reach maturity. NFTs exhibit several characteristics commonly found in early-stage financial markets. One such attribute is inefficiency, as the market is still in the process of developing a reliable and comprehensive pricing model. Additionally, there is a high degree of volatility, which can be attributed to the limited historical data and the experimental nature of NFT valuation methodologies. Market participants are also learning to navigate the complexities of NFT ownership, intellectual property rights, and the evolving regulatory landscape. The NFT market is moreover susceptible to hype cycles and speculative bubbles, similar to other emerging financial markets. These factors can result in significant price fluctuations and uncertainty, potentially creating both opportunities and risks for investors. As the market matures, it is expected that more sophisticated tools and frameworks will be developed to address these challenges, leading to increased efficiency and stability.

As the NFT market continues to evolve, one key solution will be the establishment of a balanced regulatory framework. This is essential to ensure the long-term sustainability and growth of the market, as well as to strike a balance between attracting new investors who seek greater regulation and retaining existing participants who favor the decentralized nature of blockchain technology. As of now, countries like France, Germany, and Switzerland have taken initial steps towards implementing NFT regulations in Europe, demonstrating the increasing recognition of the need for a legal framework to govern this nascent market. A well-crafted regulatory environment can provide legal clarity, enhance investor protection, and facilitate the entry of institutional investors, while also preserving the innovative spirit and autonomy that underpin the appeal of NFTs. By fostering a regulatory landscape that is both supportive and flexible, governments and regulators can help ensure the continued growth and maturation of the NFT market, while also addressing potential risks and challenges that may arise in the future.

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Table 1. Main NFT Categories. This table presents the different categories of Non-Fungible Tokens (NFTs) available in the market, along with their corresponding properties and use cases. Adapted from NonFungible.com

NFT Category	Properties	Use Cases
Art	Unique depiction with artistic function	ArtBlocks SuperRare The Currency
Collectibles	Multimedia collections. Variation of the same image, video, etc.	CryptoPunks BAYC Azuki
Gaming	Ownership of in-game assets	Axie Infinity NBA Top Shot CryptoKitties
Utilities	Utility of use in the real or digital world through the blockchain	VeeFriends ENS Nouns
Web3 & Metaverse	Expendable assets in a virtual universe, accessible through digital systems	Decentraland The Sandbox Cryptovoxels

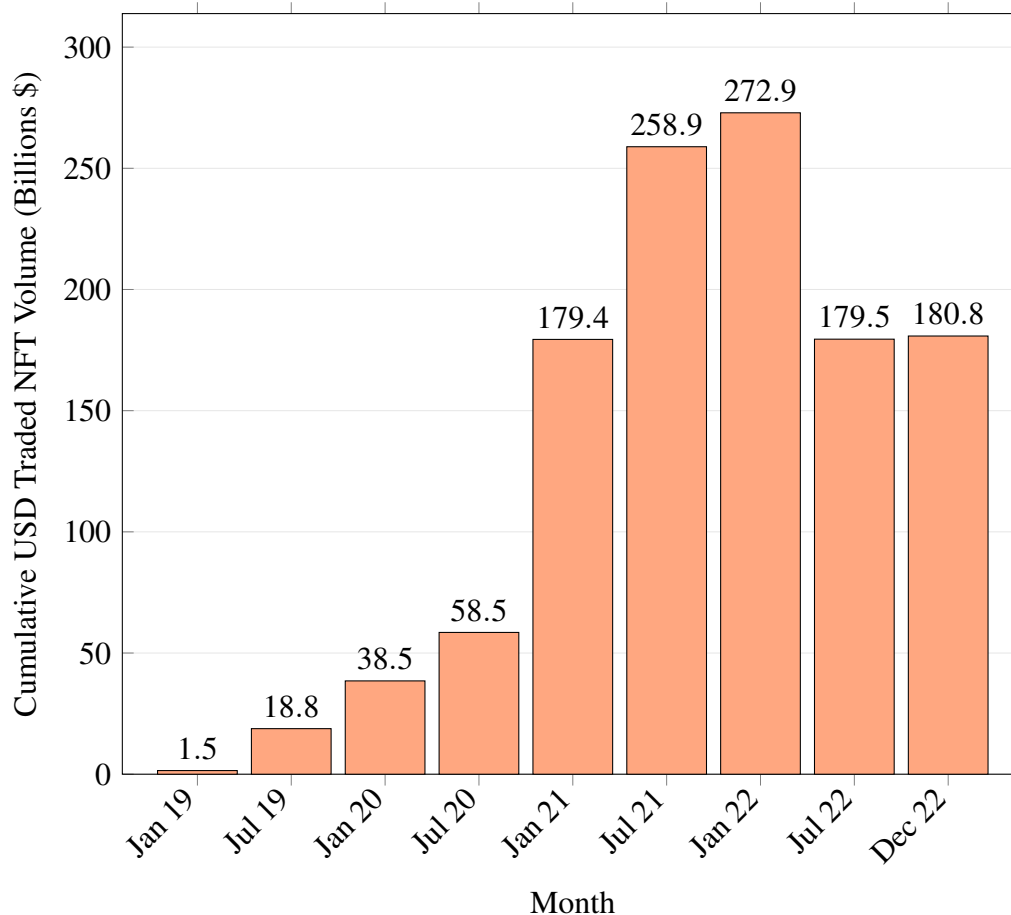


Figure 1. Cumulative Traded NFT Volume in USD. This figure illustrates the rapid growth and subsequent stabilization of the cumulative USD traded NFT volume between January 2019 and December 2022, highlighting the increased interest and investment in the NFT market. Adapted from NonFungible.com.

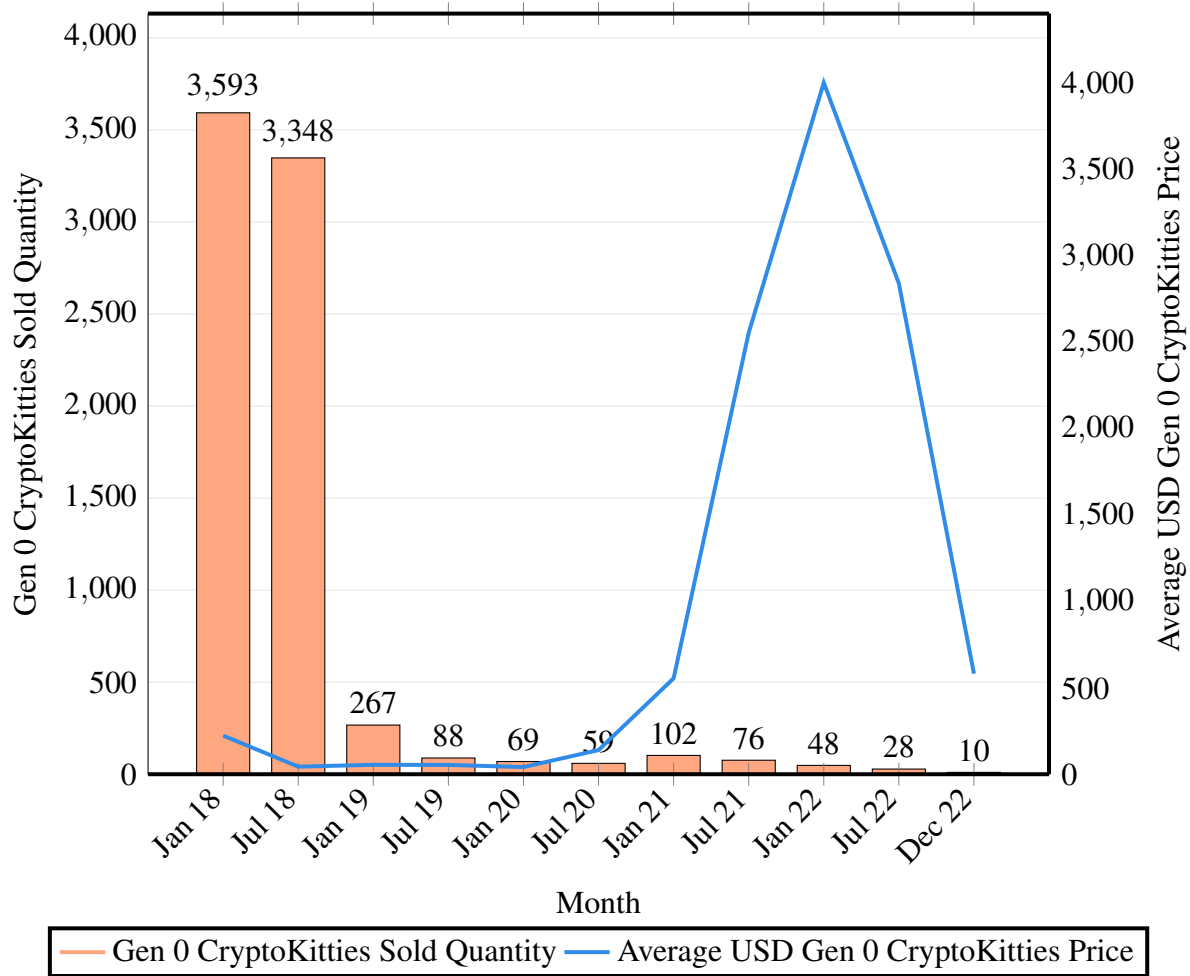


Figure 2. Gen. 0 CryptoKitties Sales. This figure illustrates the relationship between the volume and price of Generation 0 (Gen 0) CryptoKitties from January 2018 to December 2022. The graph displays an inverse relationship between the number of sales (NFT Sold Quantity) and the average price in USD, highlighting a significant decline in sales volume after October 2018. This inverse correlation is consistent with the principles of asset pricing, where scarcity plays a critical role in the valuation of non-fungible objects, such as art or collectibles. Adapted from Dune Analytics.

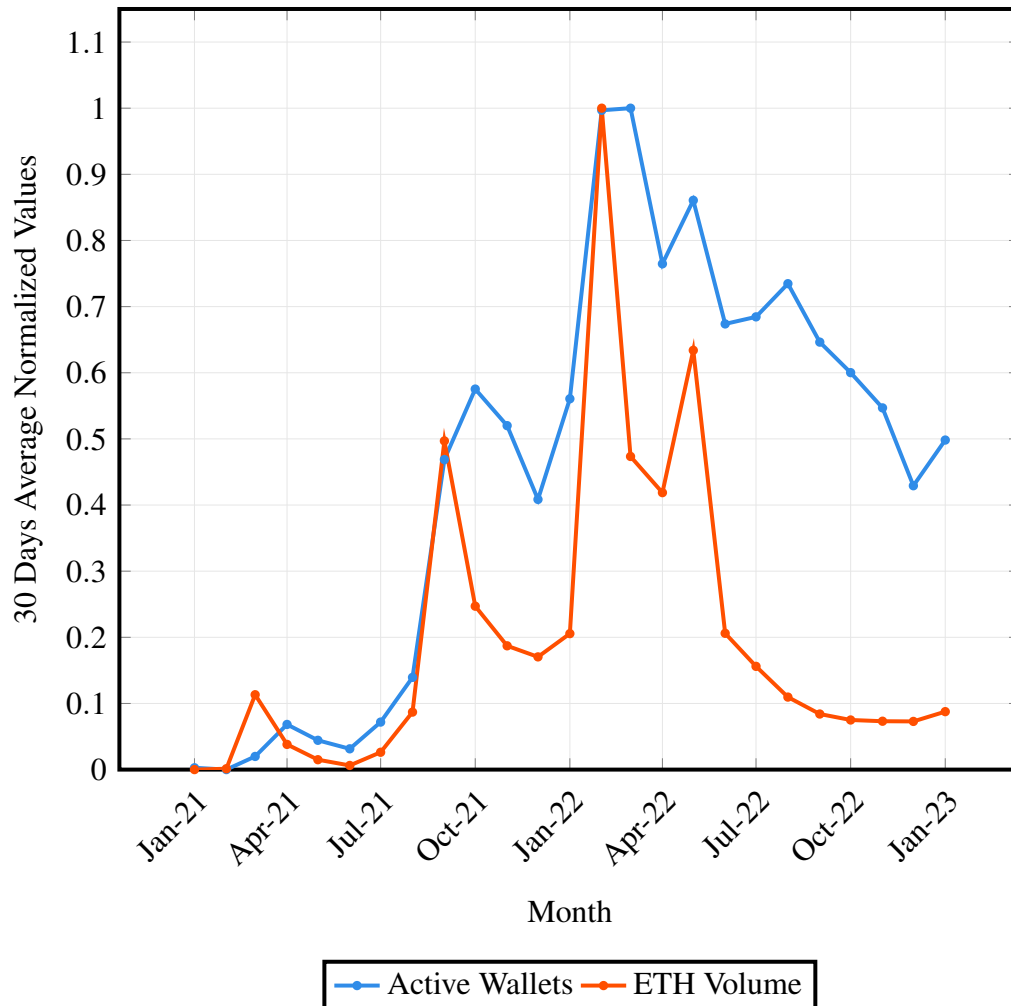


Figure 3. Normalized NFT ETH Volume and Number of Unique Active Wallets. This figure illustrates the fluctuations in the 30-day average normalized values of active wallets (blue) and ETH volume (orange) over the two-year period from January 2021 to January 2023. Both indicators exhibit an overall upward trend, with a significant surge in activity between August and September 2021. Active wallets experienced another peak in March 2022 before gradually declining, while ETH volume reached its peak in February 2022 and has since stabilized at a lower level. Adapted from Dune Analytics.